

Methodology & Data Dictionary

Food Security Survey 2025

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Dataset methodology guide for portal testing

Prepared for research, monitoring, evaluation, and knowledge-sharing workflows

Version 1.0 | October 2026

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Dataset at a glance

This guide accompanies the sample CSV dataset titled *Mozambique Household Food Security & Crop Yield Survey (2025)*. It describes the intended sampling logic, field protocol, variable definitions, coding conventions, and responsible-use notes.

1 Executive summary

The Mozambique Household Food Security & Crop Yield Survey (2025) is a structured sample dataset designed to support testing of the CE-AFSN dataset portal. It represents household-level observations from four provinces: Maputo, Gaza, Inhambane, and Sofala. The sample includes household composition, primary crop, estimated harvest yield, dietary diversity, and a derived food-insecurity status.

The intended survey design is a province-stratified, cross-sectional household survey. Districts are selected to provide geographic and agro-ecological variation, while households are selected within communities using a documented field listing and systematic selection procedure. The sample is suitable for demonstrating search, filtering, metadata, download, and visualization features; it is not intended to produce official provincial or national estimates without survey weights, full sampling documentation, and an approved analysis plan.

Important use note

This is a realistic synthetic sample for portal testing. It must not be interpreted as verified official statistics or as evidence of conditions in any named household or community.

2 Sampling methodology

2.1 Design and geographic coverage

The reference design uses four provincial strata: Maputo, Gaza, Inhambane, and Sofala. Districts are selected purposively for testing coverage across southern and central Mozambique and to represent a mix of irrigated, rain-fed, peri-urban, coastal, and inland production contexts. Within each selected district, enumeration areas or communities would be listed before household selection.

2.2 Household selection

A recommended operational sequence is:

1. Obtain or prepare a community and household listing for each selected enumeration area.
2. Assign a unique household number that is not derived from a person's name or national identifier.
3. Select households using a fixed interval after a random start, or another protocol approved in the field manual.
4. Record non-response, replacement, and eligibility decisions in a separate fieldwork log.
5. Apply survey weights only after the probability of selection and non-response adjustments have been documented.

2.3 Sample interpretation

The portal sample contains 18 records and is intentionally balanced across the four provinces for testing. A production survey should define the target population, minimum sample size, precision requirements, cluster design, replacement rules, and weighting approach before data collection begins.

3 Data collection protocol

Data collection is assumed to take place during or shortly after the 2025 main agricultural season. Trained enumerators would use a standardized questionnaire in the locally appropriate language, with a tablet-based form or controlled paper-to-digital workflow. Supervisors would review submissions daily for completeness, range errors, duplicate household identifiers, and unusual crop-yield values.

The protocol should include the following safeguards:

- Obtain informed consent before the interview.
- Interview an adult household member who is knowledgeable about food access and agricultural production.
- Use a consistent recall period for the dietary diversity questions.
- Record harvest yield using a common unit and document whether the value is measured, farmer-reported, or estimated.
- Remove direct personal identifiers from the research dataset.
- Encrypt devices and transfer files through an approved secure channel.
- Escalate safeguarding concerns or requests for respondent assistance through the approved referral pathway.

3.1 Provincial field notes

Province Example districts Fieldwork consideration		
Maputo	Boane; Marracuene; Manhiça; Moamba	Peri-urban and smallholder contexts; confirm access and seasonal labor patterns.
Gaza	Chókwè; Guijá; Massingir; Xai-Xai; Chibuto	Irrigation and drought exposure may affect crop yield and food access.
Inhambane	Maxixe; Inharrime; Vilankulo; Morrumbene	Include coastal and inland communities; plan for travel time and dispersed settlements.
Sofala	Dondo; Nhamatanda; Búzi; Marromeu; Beira	Consider flood exposure, transport interruptions, and urban-rural differences.

4 Data dictionary

The fields below define the CSV structure. Values should remain consistent across portal imports, downloads, and API responses.

Field Definition Allowed values / rules	Type		
Household_ID	Text	Unique anonymized identifier for the household record.	Format: HH-2025-xxx.
Province	Categorical	Province in which the household was sampled.	Maputo; Gaza; Inhambane; Sofala.
District	Text	Administrative district where the household was sampled.	Use the district name consistently.
Household_Size	Integer	Number of people who usually live and eat from the household.	Positive whole number; do not include visitors temporarily present.

Field	Type	Definition	Allowed values / rules
Primary_Crop	Categorical	Crop contributing the largest share of the household's reported harvest.	Maize; Cassava; Beans; Sweet Potato.
Harvest_Yield_Kg	Numeric	Harvest quantity for the primary crop, recorded in kilograms.	Non-negative number; retain decimals where measured or estimated.
HDDS_Score	Integer	Household Dietary Diversity Score based on the number of food groups consumed in the reference period.	Integer from 1 to 12 in this sample.
Food_Insecure_Status	Binary categorical	Portal-ready classification indicating whether the household is flagged as food insecure.	Yes or No; document the classification rule in the analysis plan.

5 Quality assurance and processing notes

5.1 Validation checks

Before publication to a dataset portal, run the following checks:

- Confirm that each Household_ID is unique and matches the expected pattern.
- Confirm that province values are restricted to the four approved categories.
- Confirm that Household_Size is an integer greater than zero.
- Confirm that Harvest_Yield_Kg is numeric and non-negative.
- Confirm that HDDS_Score is an integer from 1 through 12.
- Confirm that Food_Insecure_Status is exactly Yes or No.
- Check that every district belongs to the stated province and that no direct identifiers are present.

5.2 Missing data and revisions

Missing values should be represented consistently, preferably as blank cells or a documented null value rather than mixed strings such as N/A, unknown, and -. Any correction to a released dataset should increment the version number, record the change in a changelog, and preserve the original file where governance rules require version history.

5.3 Responsible interpretation

HDDS is a dietary diversity indicator and should not be treated as a complete measure of household food security on its own. The binary food-insecurity status is a classification field for portal demonstration and requires a documented derivation rule in a production release. Yield values may reflect measurement, recall, or estimation and should be interpreted together with crop area, season, weather, input use, and measurement-method metadata when those variables are available.

6 Contact and stewardship

For questions about the sample dataset, methodology guide, metadata, or portal testing workflow, contact:

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Document status: Sample methodology guide for dataset portal testing. This document should be reviewed and approved by the CE-AFSN data steward before use with production data.